

Chapter - 2

Higher Order Derivatives

2.1 Introduction:

If $y = f(x)$ is differentiable in an interval, then its derivative with respect to x is $f'(x)$, y' or $\frac{dy}{dx}$. If $f'(x)$ is also differentiable in the interval, then its derivative is given by $f''(x)$, y'' or $\frac{d^2y}{dx^2}$.

Similarly, if the $f(x)$ is differentiable in the interval n times, then its n^{th} derivative is $f^{(n)}(x)$, $y^{(n)}$ or $\frac{d^ny}{dx^n}$.

If $y = f(x)$, then the *higher order derivatives* are denoted by

$\frac{dy}{dx}$,	$f'(x)$,	y_1 ,	Dy ,	$\frac{d}{dx} f(x)$
$\frac{d^2y}{dx^2}$,	$f''(x)$,	y_2 ,	D^2y ,	$\frac{d^2}{dx^2} f(x)$
$\frac{d^3y}{dx^3}$,	$f'''(x)$,	y_3 ,	D^3y ,	$\frac{d^3}{dx^3} f(x)$
:	:	:	:	:
$\frac{d^ny}{dx^n}$,	$f^{(n)}(x)$,	y_n ,	D^ny ,	$\frac{d^n}{dx^n} f(x)$

2.2. The n^{th} derivatives of some special functions

The n^{th} derivatives of the following functions are to be obtained by the method of induction.

1. If $y = x^m$, then $y_n = \frac{m!}{(m-n)!} x^{m-n}$ where m is positive integer, $m > n$.

Here $y = x^m$

Differentiating it n times successively,

$$y_1 = mx^{m-1}$$

$$\text{or } y_2 = m(m-1)x^{m-2}$$

$$\text{or } y_3 = m(m-1)(m-2)x^{m-3}$$

$$\text{or } y_4 = m(m-1)(m-3)x^{m-4}$$

$$\vdots$$

$$\vdots$$

$$\text{or } y_n = m(m-1)(m-2)(m-3)\dots(m-n+1)x^{m-n}$$

$$\text{or } y_n = \frac{m(m-1)(m-2)\dots(m-n+1)(m-n)\dots 3.2.1}{(m-n)\dots\dots\dots 3.2.1} x^{m-n}, m > n.$$

$$\therefore y_n = \frac{m!}{(m-n)!} x^{m-n}$$

Cor.

When $m = n$

$$y_n = n!$$

For example,

$$\text{If } y = x^3, \text{ then } y_3 = 3!$$

$$\text{For } y = x^3$$

Differentiating it successively,

$$y_1 = 3x^2$$

$$\text{or } y_2 = 3 \times 2x$$

$$\text{or } y_3 = 3 \times 2 \times 1$$

$$\therefore y_3 = 3!$$

2. If $y = (ax + b)^m$, then $y_n = a^n \frac{m!}{(m-n)!} (ax + b)^{m-n}$ where m is positive integer, $m > n$.

Here $y = (ax + b)^m$

Differentiating it successively,

$$y_1 = a.m (ax + b)^{m-1}$$

$$y_2 = a^2 m(m-1) (ax + b)^{m-2}$$

$$y_3 = a^3 m(m-1)(m-2) (ax + b)^{m-3}$$

$$\vdots$$

$$y_n = a^n m(m-1)(m-2)\dots(m-n+1)(ax + b)^{m-n}$$

$$\therefore y_n = a^n \frac{m!}{(m-n)!} (ax + b)^{m-n}$$

3. If $y = e^{ax}$, then $y_n = a^n e^{ax}$

Here $y = e^{ax}$

Differentiating it successively,

$$y_1 = ae^{ax}$$

$$\text{or } y_2 = a^2 e^{ax}$$

$$\text{or } y_3 = a^3 e^{ax}$$

$$\vdots$$

$$\vdots$$

$$\therefore y_n = a^n e^{ax}$$

4. If $y = \frac{1}{(ax+b)}$, then $y_n = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$

Here $y = \frac{1}{(ax+b)}$

Differentiating it successively,

$$y_1 = (-1) \frac{a \cdot 1}{(ax+b)^2}$$

$$\text{or } y_2 = (-1)^2 \frac{a^2 \cdot 1 \cdot 2}{(ax+b)^3}$$

$$\text{or } y_3 = (-1)^3 \frac{a^3 \cdot 1 \cdot 2 \cdot 3}{(ax+b)^4}$$

$$\vdots$$

$$\therefore y_n = (-1)^n \frac{a^n \cdot n!}{(ax+b)^{n+1}}$$

Cor. 1.

$$\text{If } y = \frac{1}{x}, \text{ then } y_n = \frac{(-1)^n n!}{x^{n+1}}$$

Cor. 2.

$$\text{If } y = \frac{1}{x+a}, \text{ then } y_n = \frac{(-1)^n n!}{(x+a)^{n+1}}$$

$$\text{Cor. 3. If } y = \frac{1}{x-a}, \text{ then } y_n = \frac{(-1)^n n!}{(x-a)^{n+1}}$$

5. If $y = \sin(ax + b)$, then $y_n = a^n \sin \left(ax + b + \frac{n\pi}{2} \right)$.

Here $y = \sin(ax + b)$

Differentiating it successively,

$$y_1 = a \cos(ax + b) = a \sin \left(1 \cdot \frac{\pi}{2} + ax + b \right)$$

$$y_2 = -a^2 \sin(ax + b) = a^2 \sin \left(2 \cdot \frac{\pi}{2} + ax + b \right)$$

$$y_3 = -a^3 \cos(ax + b) = a^3 \sin \left(3 \cdot \frac{\pi}{2} + ax + b \right)$$

$\vdots$
 $\vdots$

$$\therefore y_n = a^n \sin \left(\frac{n\pi}{2} + ax + b \right)$$

Cor.

Put $b = 0$, if $y = \sin ax$, then $y_n = a^n \sin \left(\frac{n\pi}{2} + ax \right)$

6. If $y = \cos(ax + b)$, then $y_n = a^n \cos \left(\frac{n\pi}{2} + ax + b \right)$

Here $y = \cos(ax + b)$

Differentiating it successively,

$$y_1 = -a \sin(ax + b) = a \cos \left(1 \cdot \frac{\pi}{2} + ax + b \right)$$

$$\text{or } y_2 = -a^2 \cos(ax + b) = a^2 \cos \left(2 \cdot \frac{\pi}{2} + ax + b \right)$$

$$\text{or } y_3 = a^3 \sin(ax + b) = a^3 \cos \left(3 \cdot \frac{\pi}{2} + ax + b \right)$$

$\vdots$
 $\vdots$

$$\therefore y_n = a^n \cos \left(\frac{n\pi}{2} + ax + b \right)$$

Cor.

Put $b = 0$, if $y = \cos ax$, then $y_n = a^n \cos \left(\frac{n\pi}{2} + ax \right)$

7. If $y = e^{ax} \sin bx$, then $y_n = (a^2 + b^2)^{n/2} e^{ax} \sin \left(bx + n \tan^{-1} \frac{b}{a} \right)$

Here $y = e^{ax} \sin bx$

Differentiating it with respect to x ,

$$y_1 = be^{ax} \cos bx + ae^{ax} \sin bx$$

or $y_1 = e^{ax} (a \sin bx + b \cos bx)$

Put $a = r \cos \theta$, $b = r \sin \theta$, so that, $a^2 + b^2 = r^2$ and $\tan \theta = \frac{b}{a}$

Thus

$$y_1 = e^{ax} (r \cos \theta \sin bx + r \sin \theta \cos bx)$$

$$= r e^{ax} (\cos \theta \sin bx + \sin \theta \cos bx)$$

$$y_1 = r e^{ax} \sin (bx + \theta)$$

Differentiating it successively,

$$y_2 = r^2 e^{ax} \sin (bx + 2\theta)$$

or $y_3 = r^3 e^{ax} \sin (bx + 3\theta)$

$$\vdots$$

$$\vdots$$

$$y_n = r^n e^{ax} \sin (bx + n\theta)$$

$$\therefore y_n = (a^2 + b^2)^{n/2} e^{ax} \sin \left(bx + n \tan^{-1} \frac{b}{a} \right)$$

8. If $y = e^{ax} \cos bx$, then $y_n = (a^2 + b^2)^{n/2} e^{ax} \cos \left(bx + n \tan^{-1} \frac{b}{a} \right)$

We have $y = e^{ax} \cos bx$

Differentiating it with respect to x , we get

$$y_1 = -b e^{ax} \sin bx + a e^{ax} \cos bx$$

or $y_1 = e^{ax} (a \cos bx - b \sin bx)$

Put $a = r \cos \theta$, $b = r \sin \theta$ so that $a^2 + b^2 = r^2$ and $\tan \theta = \frac{b}{a}$

Thus

$$y_1 = e^{ax} (r \cos \theta \cos bx - r \sin \theta \sin bx)$$

$$= r e^{ax} (\cos \theta \cos bx - \sin \theta \sin bx)$$

$$y_1 = r e^{ax} \cos (bx + \theta)$$

Differentiating it successively,

$$y_2 = r^2 e^{ax} \cos (bx + 2\theta)$$

$$\text{or } y_3 = r^3 e^{ax} \cos (bx + 3\theta).$$

$$\vdots$$

$$\text{or } y_n = r^n e^{ax} \cos (bx + n\theta)$$

$$\therefore y_n = (a^2 + b^2)^{n/2} e^{ax} \cos \left(bx + n \tan^{-1} \frac{b}{a} \right)$$

9. If $y = \frac{1}{x^2 - a^2}$, then $y_n = \frac{(-1)^n n!}{2a} \left[\frac{1}{(x-a)^{n+1}} - \frac{1}{(x+a)^{n+1}} \right]$

Here $y = \frac{1}{x^2 - a^2}$

or $y = \frac{1}{(x-a)(x+a)}$

or $y = \frac{1}{2a} \left[\frac{1}{x-a} - \frac{1}{x+a} \right]$

Thus

$$y_n = \frac{1}{2a} \left[\frac{(-1)^n n!}{(x-a)^{n+1}} - \frac{(-1)^n n!}{(x+a)^{n+1}} \right]$$

$$\therefore y_n = \frac{(-1)^n n!}{2a} \left[\frac{1}{(x-a)^{n+1}} - \frac{1}{(x+a)^{n+1}} \right]$$

10. If $y = \frac{1}{x^2 + a^2}$, then $y_n = \frac{(-1)^n n!}{a^{n+2}} \sin^{n+1} \left(\tan^{-1} \frac{a}{x} \right) \sin \left\{ (n+1) \tan^{-1} \frac{a}{x} \right\}$

Here $y = \frac{1}{x^2 + a^2}$

or $y = \frac{1}{x^2 - i^2 a^2} = \frac{1}{(x+ia)(x-ia)}$

$$= \frac{1}{2ai} \left[\frac{1}{x-ia} - \frac{1}{x+ia} \right]$$

$$y_n = \frac{1}{2ai} \left[\frac{(-1)^n n!}{(x-ia)^{n+1}} - \frac{(-1)^n n!}{(x+ia)^{n+1}} \right]$$

$$= \frac{(-1)^n n!}{2ai} \left[(x-ia)^{-(n+1)} - (x+ia)^{-(n+1)} \right]$$

Put $x = r \cos \theta$, $a = r \sin \theta$, so that, $x^2 + a^2 = r^2$, $\tan \theta = \frac{a}{x}$

Thus

$$\begin{aligned}
 y_n &= \frac{(-1)^n n!}{2ai} [(r \cos \theta - i r \sin \theta)^{-(n+1)} - (r \cos \theta + i r \sin \theta)^{-(n+1)}] \\
 &= \frac{(-1)^n n!}{2ai} r^{-(n+1)} [\cos(n+1)\theta + i \sin(n+1)\theta - \cos(n+1)\theta \\
 &\quad + i \sin(n+1)\theta] \\
 &= \frac{(-1)^n n!}{2ai r^{n+1}} 2i \sin(n+1)\theta = \frac{(-1)^n n!}{a r^{n+1}} \sin(n+1)\theta \\
 &= \frac{(-1)^n n!}{a^{n+2}} \sin^{n+1} \theta \sin(n+1)\theta \\
 \therefore y_n &= \frac{(-1)^n n!}{a^{n+2}} \sin^{n+1} \left(\tan^{-1} \frac{a}{x} \right) \sin \left\{ (n+1) \tan^{-1} \frac{a}{x} \right\}
 \end{aligned}$$

Leibnitz's Theorem

If u and v are two function of x , then the n^{th} derivative of their product is

$$(uv)_n = u_n v + c(n, 1) u_{n-1} v_1 + c(n, 2) u_{n-2} v_2 + \dots + c(n, r) u_{n-r} v_r + \dots + uv_n$$

where the suffices of u and v denote the order of differentiation of u and v with respect to x .

Proof:

Let $y = uv$

Differentiating it with respect to x , we get

$$y_1 = u_1 v + uv_1$$

$$y_2 = u_2 v + 2u_1 v_1 + uv_2$$

$$y_3 = u_3 v + 3u_2 v_1 + 3u_1 v_2 + uv_3$$

It shows that the theorem is true for $n = 1, 2, 3$.

Let us suppose that the theorem is true for $n = m$ and need to show that it is true for $n = m + 1$.

$$\begin{aligned}
 \text{Let } y_m &= u_m v + C(m, 1) u_{m-1} v_1 + C(m, 2) u_{m-2} v_2 + \dots \\
 &\quad + C(m, r-1) u_{m-r+1} v_{r-1} + C(m, r) u_{m-r} v_r + \dots + uv_m
 \end{aligned}$$

Differentiating it with respect to x , we get

$$y_{m+1} = u_{m+1} v + u_m v_1 + C(m, 1) u_{m-1} v_1 + C(m, 1) u_{m-1} v_2$$

$$\begin{aligned}
 & + C(m, 2) u_{m-1} v_2 + C(m, 2) u_{m-2} v_3 + \dots \\
 & \dots + C(m, r-1) u_{m-r+2} v_{r-1} + C(m, r-1) u_{m-r+1} v_r \\
 & + C(m, r) u_{m-r+1} v_r + \dots + u v_{m+1} \\
 & = u_{m+1} v + [1 + C(m, 1)] u_m v_1 + [C(m, 1) + C(m, 2)] u_{m-1} v_2 \\
 & + \dots + [C(m, r-1) + C(m, r)] u_{m-r+1} v_r + \dots + u v_{m+1}
 \end{aligned}$$

Using $1 + C(m, 1) = C(m+1, 1)$,
 $C(m, 1) + C(m, 2) = C(m+1, 2)$,
 $C(m, r-1) + C(m, r) = C(m+1, r)$.

We get

$$\begin{aligned}
 y_{m+1} = & u_{m+1} v + C(m+1, 1) u_m v_1 + C(m+1, 2) u_{m-1} v_2 + \dots \\
 & \dots + C(m+1, r) u_{m-r+1} v_r + \dots + u v_{m+1}
 \end{aligned}$$

Thus, the theorem is true for $n = m + 1$ if it is true for $n = m$, and hence it is true for $n = 1, 2, 3$ and so on. It can be concluded that the theorem is true for every positive integral value of n .

Worked out Examples

Ex. 1: If $y = \sqrt{x}$ then find y_n .

Solution:

Given function is

$$y = \sqrt{x}$$

Differentiating it successively,

$$y_1 = \frac{1}{2} x^{-1/2}$$

$$\text{or } y_2 = \frac{1}{2} \left(-\frac{1}{2}\right) x^{-(1/2)-1}$$

$$\text{or } y_3 = \frac{1}{2} \left(-\frac{1}{2}\right) \left(-\frac{1}{2}-1\right) x^{-(1/2)-2}$$

$$\therefore \therefore \therefore \therefore \therefore \therefore$$

$$\text{or } y_n = \frac{1}{2} \left(-\frac{1}{2}\right) \left(-\frac{1}{2}-1\right) \dots \left(-\frac{1}{2}-n+2\right) x^{-(1/2)-n+1}$$

$$\therefore y_n = \frac{(-1)^{n-1}}{2^n} [1.3.5 \dots (2n-3)] x^{-(1/2)-n}$$

Ex. 2: If $y = \frac{x^n}{x+1}$, then find y_n .

Solution:

Here $y = \frac{x^n}{x+1}$

By actual division

$$y = x^{n-1} - x^{n-2} + x^{n-3} \dots + (-1)^n \frac{1}{x+1}$$

Differentiating it with respect to x n times,

$$y_n = 0 + (-1)^n D^n \left(\frac{1}{x+1} \right)$$

$$\text{or } y_n = (-1)^n \frac{(-1)^n n!}{(x+1)^{n+1}}$$

$$\text{or } y_n = (-1)^{2n} \frac{n!}{(x+1)^{n+1}}$$

$$\therefore y_n = \frac{n!}{(x+1)^{n+1}}$$

Handwritten notes showing polynomial division:

$$\begin{array}{r} x^3 \\ x+1 \overline{) x^3 + 0x^2 + 0x + 0} \\ \underline{x^2 + x} \\ x^2 + 0x + 0 \\ \underline{x^2 + x} \\ -x \\ \underline{-x-1} \\ 1 \end{array}$$

Ex. 3: If $y = (x^2-1)^n$, then $(x^2-1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0$ 2065 B.E.

Solution:

Here $y = (x^2-1)^n$

Differentiating it with respect to x , we get

$$y_1 = n(x^2-1)^{n-1} \cdot 2x$$

$$\text{or } y_1 = \frac{n(x^2-1)^n}{(x^2-1)} \cdot 2x = \frac{ny}{x^2-1} \cdot 2x$$

$$\text{or } (x^2-1)y_1 = 2nxy$$

Again, differentiating it with respect to x , we get

$$(x^2-1)y_2 + 2xy_1 = 2nxy_1 + 2ny$$

$$(x^2-1)y_2 + (2x-2nx)y_1 = 2ny$$

Differentiating it n times using Leibnitz's theorem

$$(x^2-1)y_{n+2} + \frac{n}{1!} y_{n+1} \cdot 2x + \frac{n(n-1)}{2!} y_n \cdot 2 + (2x-2nx)y_{n+1}$$

$$+ \frac{n}{1!} y_n (2-2n) = 2ny_n$$

$$\text{or } (x^2-1)y_{n+2} + (2nx+2x-2nx)y_{n+1} = 2ny_n$$

$$+ (n^2 - n + 2n - 2n^2 - 2n) y_n = 0$$

$$\therefore (x^2 - 1) y_{n+2} + 2xy_{n+1} - n(n+1) y_n = 0$$

Ex. 4: If $y = \sin mx + \cos mx$, then prove $y_n = m^n \{1 + (-1)^n \sin 2mx\}^{1/2}$

Solution:

Given function is

$$y = \sin mx + \cos mx$$

Using the formula for n^{th} derivative

$$\begin{aligned} y_n &= m^n \left\{ \sin \left(\frac{n\pi}{2} + mx \right) + \cos \left(\frac{n\pi}{2} + mx \right) \right\} \\ &= m^n \left[\left\{ \sin \left(\frac{n\pi}{2} + mx \right) + \cos \left(\frac{n\pi}{2} + mx \right) \right\}^2 \right]^{1/2} \\ &= m^n \left\{ 1 + \sin \left(\frac{n\pi}{2} + mx \right) \cos \left(\frac{n\pi}{2} + mx \right) \right\}^{1/2} \\ &= m^n [1 + \sin (n\pi + 2mx)]^{1/2} \end{aligned}$$

$$\text{or } y_n = m^n [1 + \sin 2mx \cos n\pi + \cos 2mx \sin n\pi]^{1/2}$$

Using $\sin n\pi = 0$, $\cos n\pi = (-1)^n$

$$y_n = m^n [1 + (-1)^n \sin 2mx]^{1/2}$$

Ex. 5: If $y = (x + \sqrt{1+x^2})^m$, then show that $(1+x^2) y_2 + xy_1 - m^2 y = 0$ and hence prove that $(1+x^2) y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2) y_n = 0$

Solution:

Given function is

$$y = (x + \sqrt{1+x^2})^m \quad \dots\dots\dots(1)$$

Differentiating it with respect to x

$$y_1 = m(x + \sqrt{1+x^2})^{m-1} \left(1 + \frac{1}{2\sqrt{1+x^2}} 2x \right)$$

$$\text{or } y_1 = m(x + \sqrt{1+x^2})^{m-1} \frac{(x + \sqrt{1+x^2})}{\sqrt{1+x^2}}$$

$$\text{or } y_1 = \frac{m(x + \sqrt{1+x^2})^m}{\sqrt{1+x^2}} = \frac{my}{\sqrt{1+x^2}} \quad [\text{from (1)}]$$

$$\text{or } (1+x^2) y_1^2 = m^2 y^2$$

Again differentiating it

$$(1+x^2) 2y_1 y_2 + 2xy_1^2 = 2ym^2 y_1$$

$$\text{or } (1+x^2) y_2 + xy_1 - m^2 y = 0 \quad \dots\dots\dots(2)$$

Differentiating it n times by using Leibnitz's theorem

$$(1+x^2)y_{n+2} + \frac{n}{1!}y_{n+1}2x + \frac{n(n-1)}{2!}2 + xy^{n+1} + \frac{n}{1!}y_n - m^2y_n = 0$$

$$\therefore (1+x^2)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0$$

Ex. 6: If $y^{1/m} + y^{-1/m} = 2x$, show that

$$(x^2 - 1)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0$$

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Solution:

Given function is

$$y^{1/m} + y^{-1/m} = 2x$$

Differentiating it with respect to x ,

$$\frac{1}{m}y^{(1/m)-1}y_1 + \frac{(-1)}{m}y^{-(1/m)-1}y_1 = 2$$

$$\text{or } (y^{1/m} - y^{-1/m})y_1 = 2my$$

Squaring,

$$(y^{1/m} - y^{-1/m})^2 y_1^2 = 4m^2 y^2$$

$$\text{or } [(y^{1/m} + y^{-1/m})^2 - 4] y_1^2 = 4m^2 y^2,$$

$$\text{or } (4x^2 - 4)y_1^2 = 4m^2 y^2$$

$$\text{or } (x^2 - 1)y_1^2 = m^2 y^2$$

Again differentiating with respect to x ,

$$(x^2 - 1)2y_1 y_2 + 2x \cdot y_1^2 = 2m^2 y y_1$$

$$\text{or } (x^2 - 1)y_2 + xy_1 - m^2 y = 0$$

Differentiating it n times by using Leibnitz's theorem

$$(x^2 - 1)y_{n+2} + \frac{n}{1!}2x \cdot y_{n+1} + \frac{n(n-1)}{2!}2 \cdot y_n + xy_{n+1} + \frac{n}{1!}y_n - m^2 y_n = 0$$

$$(x^2 - 1)y_{n+2} + (2nx + x)y_{n+1} + (n^2 - n^2 + n^2 - m^2)y_n = 0$$

$$\therefore (x^2 - 1)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0$$

Ex. 7: If $y = \log(x + \sqrt{a^2 + x^2})$, then show that $(a^2 + x^2)y_2 + xy_1 = 0$
and hence show that $(a^2 + x^2)y_{n+2} + (2n+1)xy_{n+1} + n^2 y_n = 0$

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Solution:

Given function is

$$y = \log(x + \sqrt{a^2 + x^2})$$

Differentiating it with respect to x ,

$$y_1 = \frac{1}{(x + \sqrt{a^2 + x^2})} \left(1 + \frac{1}{2} \cdot \frac{2x}{\sqrt{a^2 + x^2}} \right)$$

or $y_1 = \frac{1}{\sqrt{a^2 + x^2}}$

Again differentiating it with respect to x

$$y_2 = -\frac{1}{2(a^2 + x^2)^{3/2}} \cdot 2x,$$

or $(a^2 + x^2) y_2 = -\frac{x}{\sqrt{a^2 + x^2}} = -xy_1$

or $(a^2 + x^2) y_2 + xy_1 = 0$

Using Leibnitz's theorem

$$(a^2 + x^2) y_{n+2} + \frac{n}{1!} 2x \cdot y_{n+1} + \frac{n(n-1)}{2!} 2 \cdot y_n + xy_{n+1} + \frac{n}{1!} y_n = 0$$

$$\therefore (a^2 + x^2) y_{n+2} + (2n+1)xy_{n+1} + n^2 y_n = 0.$$

Exercise-2

1. Find the n^{th} derivative of the following:

i. $y = (a - bx)^m$

ii. $y = (2 - 3x)^n$

iii. $y = \frac{1}{\sqrt{x}}$

iv. $y = x^{2n}$

v. $y = \frac{1}{a-x}$

vi. $y = \frac{x^n}{x-1}$

vii. $y = \frac{x}{a+bx}$

viii. $y = \frac{x}{x^2+a^2}$

(ix) $y = \tan^{-1} \left(\frac{1+x}{1-x} \right)$

(x) $y = \tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$

2. If $y = \frac{1}{x^n}$, then show that $y_1(1) = 0$

3. Find y_n if i. $y = e^{ax} x^3$ ii. $y = x^3 \log x$

4. If $y = \tan^{-1} x$, then prove that $(1-x^2) y_{n+2} - 2nxy_{n+1} + n(n-1) y_n = 0$

5. If $y = e^{a \tan^{-1} x}$, then prove that
 $(1 + x^2) y_{n+2} + (2nx + 2x - a) y_{n+1} + n(n+1) y_n = 0$ 2061/067 B.E.
6. If $y = ae^{mx} + be^{-mx}$, then prove that $y_2 - m^2 y = 0$
7. If $y = e^{ax} \sin bx$, then prove that
 $y_{n+1} = 2a y_n - (a^2 + b^2) y_{n-1}$
8. If $y = x^{n-1} \log x$, then prove that $xy_n = (n-1)!$
9. If $y = a \cos(\log x) + b \sin(\log x)$, then prove that
 $x^2 y_{n+1} + (2n+1) xy_{n+1} + (n^2 + 1) y_n = 0$ 2069 B.E.
10. If $y = \sin^{-1} x$, then prove that
 $(1 - x^2) y_{n+2} - (2n+1) xy_{n+1} - n^2 y_n = 0$ 2068 B.E.
11. If $y = e^{a \sin^{-1} x}$, then prove that
 $(1 - x^2) y_{n+2} - (2n+1) xy_{n+1} - (n^2 + a^2) y_n = 0$
12. If $y = e^{x^2}$, then show that $y_{n+1} - 2xy_n - 2ny_{n-1} = 0$
13. If $y = (\sin^{-1} x)^2$, then show that
 $(1 - x^2) y_{n+2} - (2n+1) xy_{n+1} - n^2 y_n = 0$ 2072 Kartik B.E.
14. If $y = \sin(m \sin^{-1} x)$, then show that
 $(1 - x^2) y_{n+2} - (2n+1) xy_{n+1} + (m^2 - n^2) y_n = 0$ 2070/071 Chaitra B.E.

Answers

1. i. $\frac{m! (-1)^n b^n}{(m-n)!} (a-bx)^{m-n}$ ii. $(-1)^n 3^n n!$
- iii. $(-1)^n \frac{1.3.5 \dots (2n-1)}{2^n x^{n+\frac{1}{2}}}$ iv. $2^n \{1.3.5 \dots (2n-1)\} x^n$
- v. $\frac{n!}{(a-x)^{n+1}}$ vi. $\frac{(-1)^n n!}{(x-1)^{n+1}}$

vii. $\frac{(-1)^{n+1} a \cdot b^{n-1} n!}{(a+bx)^{n+1}}$

viii. $\frac{(-1)^n n!}{a^{n+1}} \sin^{n+1} \theta \cos(n+1) \theta$, where $\theta = \tan^{-1} \frac{a}{x}$.

ix. $(-1)^{n-1} (n-1)! \sin^n \theta \sin n\theta$, where $\theta = \tan^{-1} \left(\frac{1}{x} \right)$

x. $(-1)^{n-1} \frac{1}{2} (n-1)! \sin^n \theta \sin \theta$ where $\theta = \tan^{-1} \left(\frac{1}{x} \right)$

3. i. $e^{ax} a^{n-3} \{a^3 x^3 + 3na^2 x^2 + 3n(n-1) ax + n(n-1)(n-2)\}$.

ii. $(-1)^n \frac{6(n-4)!}{x^{n-3}}$

